Studying the Cross-Domain Aptitude of State-of-the-Art Legal Summarization Algorithms

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by

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April, 2023

Declaration

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Abstract

Text summarization in the legal domain has been a long standing challenge due to the lack of large datasets and the long nature of legal proceedings which are filled with jargon. Hence, unsupervised models are very common because they need no training data. However, with the advent of few-shot learning, many supervised models have been proposed to tackle this challenge. However, most of these models are invariably trained on data from a particular context, i.e., on the documents from a certain country. In this study, we show that supervised models perform better in an out-of-context setting than a standalone unsupervised approach. We further investigate the utility of domain adaptation in improving the performance of these models.

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Chapter 1

Introduction

1.1 Why is Legal Summarization Important

Law practitioners go through hundreds of past cases routinely to find relevant cases to cite as precedents for an ongoing case that may potentially lead to a favourable outcome. However, legal proceedings tend to be long-drawn and free-flowing with a lot of jargon. Hence, summaries play a very prominent role, and are drafted manually by legal experts, incurring huge costs both in terms of time and expertise.

1.2 What is Natural Language Processing?

Natural language refers to the way humans, communicate with each other using speech and text. Natural Language Processing (NLP), is a branch of Computer Science that helps machines comprehend human languages, both spoken and written, to emulate the competence of us humans. Life without Google's search engine, Amazon's Alexa and Apple's SIRI are almost insurmountable in the modern age.

Researchers have been working on NLP for over half a century, and it is challenging due to the very nature of natural language being ambiguous and continuously evolving. It is not easy to formalize the rules dictating the passage of information using natural languages, and is hence difficult for machines to grasp. For a long time, the methods used to study NLP problems employed shallow machine learning models and time-consuming, hand-crafted features, which led to problems of its own. The advent of Deep Learning and more specifically, word embeddings like Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014) have achieved stellar performance over traditional methods, and

modern-day NLP algorithms are able to excel in complex tasks like Machine Translation, Speech Generation and Sentiment Analysis.

1.3 What is Automatic Text Summarization?

Automatic text summarization is the task of producing a shortened version of documents whilst preserving the vital parts of the original document. It was first introduced by (Luhn, 1958) in 1958, which was used in IBM 701, the first commercial computer. They used a naive Bag-of-Words (Harris, 1954) approach. The evolution of NLP between 1990 and 2000 introduced the conversion of sentences into vectors and words into their base forms. The introduction of advanced NLP techniques such as neural word embeddings (Mikolov et al., 2013; Pennington et al., 2014) and deep learning approaches such as recurrent neural networks (RNNs) and long short-term memory (LSTM) (Hochreiter et al., 1997) observed significant progress in the automatic text summarization domain. The advent of Attention and the Transformer trampled the previous SOTAs, and most recent approaches make use of the Transformer (Beltagy et al., 2020; Lewis et al., 2019; Liu et al., 2019; Raffel et al., 2019; Shukla et al., 2022) Most modern summarization algorithms can be broadly classified as:

- Extractive Summarization: The algorithm takes pre-existing sentences from the source documents and preserves only the relevant parts in the final summary.
- Abstractive Summarization: The algorithm creates new words and phrases and encapsulates the essence of the source document into the summary.

Recently, there have been numerous contributions to both kinds of algorithms, and both supervised and unsupervised approaches have achieved promising results (Barzilay et al., 1999).

1.4 NLP in Legal Summarization

In 2021, the Supreme Court of India established an AI department to monitor the administrative efficiency of legal research, clearly indicating the utility of AI in legal systems. Legal document summarization, however, comes with its own challenges. Firstly, legal proceedings tend to be long-drawn and free-flowing, making it difficult for an algorithm to understand. Moreover, summaries are drafted by legal experts, which makes all available

annotated datasets much smaller than required by most supervised algorithms. Hence, unsupervised algorithms are commonly used, as are models fine-tuned on legal corpora after pre-training on a more generic corpus.

1.5 Contributions

Legal text summarization is resource intensive and could save a lot of time and money if done well. The availability of legal texts and their reference from all major jurisdictions is implausible, given the costs incurred in manually annotating such data by lawyers and paralegals alike. The existence of a system that can perform well across several jurisdictions is vital to the advancement of automated legal summarizers. In this study, we investigate the cross-domain performance of several summarization algorithms: supervised and unsupervised; extractive and abstractive; domain agnostic and domain specific, on the IN-Abs, UK-Abs and BVA datasets (Shukla et al., 2022; Zhong et al., 2019) . We use the stiffness and stableness metrics used in (Chen, Liu, et al., 2020) for this purpose. Further, we work towards improving the cross-domain performance of these models using the methods described in (Ganin et al., 2016).

Concretely, we make the following contributions:

- Compare the in-domain (train and test datasets belong to the same corpus) and cross-domain (train and test datasets belong different corpuses) of supervised approaches to unsupervised approaches.
- Calculate the stiffness and stableness of different models to gauge the cross-domain performance better.
- Create domain agnostic encoder representations using Gradient Reversal Layers (GRLs) (Ganin et al., 2016)

Chapter 2

Related Work

2.1 Unsupervised Models for Summarization

Extractive summarization involves selecting a subset of the sentences from the source document as the summary, based on some measure of relevance. In the absence of any training data, unsupervised approaches to summarization are widely modelled as an extractive summarization problem. Such approaches use statistical and semantic features such as term frequency, cue words, word senses and textual entailment to identify important sentences in a document for the summary. (Luhn, 1958) introduced a heuristic approach of summarizing the documents based on word frequency and position of that word. LexRank (Erkan et al., 2004) uses a graph-based approach inspired by Google's PageRank algorithm (Page et al., 1999). LSA (Gong et al., 2001) combines term frequency with Singular Value Decomposition (Klema et al., 1980), and Reduction (Jing, 2000) uses a graph-based approach to determine the most important sentences in a document. However, none of these approaches are tailored to documents from the legal-domain, which vary significantly in terms of structure and vocabulary based on jurisdiction. CaseSummarizer (Polsley et al., 2016) uses term frequency along with some legal domain specific filters to determine the relevant sections of a document. LetSum (Farzindar et al., 2004) uses the thematic structure of a document to filter out unimportant sentences and MMR (Zhong et al., 2019) uses Maximum Marginal Relevance (Carbonell et al., 1998) to rank sentences based on relevance.

2.2 Supervised Models for Summarization

The introduction of the CNN/DailyMail corpus by (Nallapati, Zhou, et al., 2016) paved the way for the application of large-scale training of Deep Neural models for summarization. (Cheng et al., 2016) use word-level CNNs to encode sentences and sentence level sequence-to-sequence model to predict which sentences to include in the summary. Subsequently, (Nallapati, Zhai, et al., 2016) proposed a different model using word-level Bi-RNNs along with a sentence level Bi-RNN for predicting which sentences should be extracted, and computed separate scores for saliency, novelty and location of the sentences. BERTSumExt (Liu et al., 2019) treats summarization as a multi-label binary classification problem of sentences in the source document, with a label of 1 indicating the inclusion of the sentence in the summary. Apart from the CNN-DailyMail corpus, text summarization has also been applied to several personal and informal stories shared on Reddit (Ouyang et al., 2017), workplace meeting summarization (Carletta et al., 2006), and medical journal summarization (Mishra et al., 2014).

For a long time, most research focused solely on extractive summarization, but deep neural architectures show promising results in abstractive summarization, which is a much more challenging task as it aims at understanding the underlying meaning of the source document, and generate a summary with novel words and sentences. One of the earliest works in abstractive summarization was the Pointer-Generator approach (See et al., 2017), which uses Pointers (Vinyals et al., 2015) to copy words from the source document. More recently, transformer-based approaches like BART (Lewis et al., 2019), PEGASUS (Zhang, Zhao, et al., 2019), BERTSumAbs (Liu et al., 2019) have yielded state-of-the-art results.

One of the biggest challenges for abstractive summarization in the legal domain is the length of legal documents. Most prior work have a limit on the number of tokens they can process (most commonly 1024). LongFormer (Beltagy et al., 2020) uses a more efficient attention mechanism that enables them to increase the token limit to 16×1024 tokens. (Bajaj et al., 2021) use an extractive-abstractive approach for long documents, with a pre-trained BART model to generate the final summary. (Gidiotis et al., 2020) use a divide-and-conquer approach to split the documents and summaries into smaller summarizable problems using sentence similarity. Legal-LED and Legal-Pegasus (Sairam, 2021a,b) which train the LongFormer and PEGASUS on the SEC litigation releases dataset (https://www.sec.gov/litigation/litreleases.htm), which contains over 2700 litigations and complaints from the US Securities and Exchange Commission.

2.3 Cross-Domain Summarization

Legal systems vary widely across countries, making it difficult to train summarization models that can generalize to different jurisdictions. Moreover, the availability of legal datasets is limited to specific countries, and some countries have illegible court proceedings, making summarization even more challenging. We follow (Chen, Liu, et al., 2020), who prepare a matrix $\mathbf{A} \in \mathbb{R}^{\mathbf{N}\times\mathbf{N}}$ for a metric m and summarizer S, where N is the number of datasets under consideration, such that A_{ij} is the value of the metric m when trained on D_i and tested on D_j. Further, stiffness ($\mathbf{r}^{\mu}$) gives the absolute cross-domain performance of S, while stableness ($\mathbf{r}^{\sigma}$) quantifies the relative performance gap between the in-domain and cross-domain performance.

$$\mathbf{r}^{\mu} = \frac{1}{N \times N} \sum_{i,j} \mathbf{A}_{ij}$$

$$\mathbf{r}^{\sigma} = \frac{1}{N \times N} \sum_{i,j} \frac{\mathbf{A}_{ij}}{\mathbf{A}_{jj}} \times 100\%,$$

2.4 Adversarial Domain Adaptation

Domain adaptation is a subfield of natural language processing that deals with the problem of adapting a model trained on one domain to perform well on a different domain. In other words, it aims to address the issue of domain shift, where the distribution of the target domain data is different from that of the source domain data. (Ganin et al., 2016) proposed Domain-Adversarial Neural Networks (DANNs) for domain adaptation. A DANN consists of three main components: a feature extractor (an encoder) G_f, a predictor (a decoder) G_y, and a domain classifier G_d. The encoder learns to extract relevant features from the input data, and the decoder makes predictions on the input data based on the extracted features. The domain classifier predicts the domain of the input data based on the same extracted features. The proposed architecture is described in Figure 2.1. During the learning stage, the aim is to minimize the prediction loss on the source domain (i.e. the annotated data), and the parameters of both the feature extractor and the predictor are thus optimized in order to minimize the empirical loss for the source domain samples. At the same time, we want to make the features f domain-invariant. To this end, we seek the parameters θ_f of the feature mapping obtained from the encoder that *maximize* the loss of the domain classifier (by making the two feature

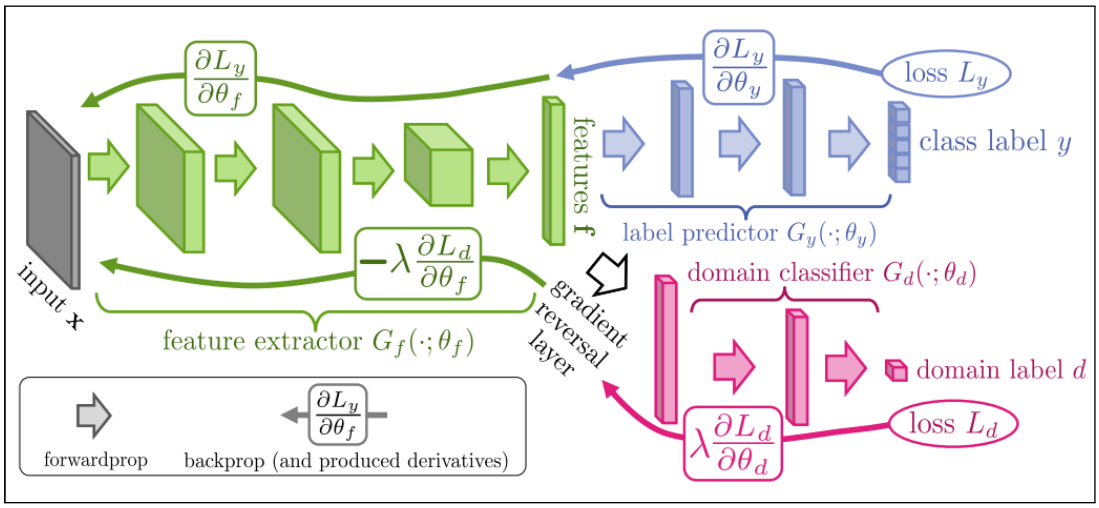

Figure 2.1: Architecture proposed by (Ganin et al., 2016). The feature extractor (green) and the predictor (blue), together form a standard feed-forward architecture. Domain adaptation is achieved by adding a domain classifier (red) connected to the feature extractor via a gradient reversal layer that multiplies the gradient by a certain negative constant during backpropagation. Otherwise, the training proceeds in a standard way and minimizes the label prediction loss (for source samples) and the domain classification loss (for all samples).

distributions as similar as possible), while simultaneously seeking parameters θ_d of the domain classifier that minimize L_d. Additionally, we seek to minimize the loss of the decoder L_y. Mathematically, the objective is:

$$E(\theta_f, \theta_y, \theta_d) = \sum_{\substack{1,\ldots,N \\ d_i=0}} L_y^i(\theta_f, \theta_y) - \lambda \sum_{1,..,N} L_d^i(\theta_f, \theta_d)$$

Note the use of the $-\lambda$ factor in the formulation. Without this factor, the optimizer will try to make the features for different domains dissimilar in order to minimize the domain classification loss. To achieve this, the authors use Gradient Reversal Layers (GRLs), which have no parameters associated with it. On a forward pass, the GRL acts as an identity mapping, but during backpropagation, it multiplies the gradient by $-\lambda$.

$$R_\lambda(\mathbf{x}) = \mathbf{x} \qquad \frac{\mathrm{d}R_\lambda}{\mathrm{d}\mathbf{x}} = -\lambda$$

(Ganin et al., 2016) use DANNs on various image classification tasks. However, there have been multiple studies that adapt DANNs for natural language processing tasks like trigger identification and title generation (Chen and Chen, 2019; Naik et al., 2020). In this study, we employ DANNs for summarization.

Chapter 3

Datasets and Metrics

3.1 Datasets for Legal Summarization

3.1.1 Board of Veterans' Appeals Dataset (BVA)

Tens of thousands of veterans apply their rejected disability benefits application to the US Board of Veterans' Appeals (BVA), which in turn affirms, denies or remands the decision. The BVA has published an average of 55,000 cases every year since 1991, most of which have been drafted by judges and their support staff. It has a huge backlog and it is very hard for the BVA and the applying veterans to be fairly heard and judged given the number of applicants. The dataset proposed by (Zhong et al., 2019) consists of 92 annotated extractive summaries, created using 6 types of single issue BVA cases related to Post Traumatic Stress Disorder (PTSD). They also propose a test set of 20 such cases, with summaries of 4 annotators. In this paper, we use the aforementioned test set for the evaluation of both extractive and abstractive models in a cross-domain setting.

3.1.2 Indian-Abstractive Dataset (IN-Abs)

The Supreme Court of India is the highest judicial authority in the country. The Indian-Abstractive (IN-Abs) dataset, proposed by (Shukla et al., 2022), consists of judgements along with abstractive summaries from the Legal Information Institute India website (http://www.liiofindia.org/in/cases/cen/INSC/), which strives to provide free access to Indian law databases. The training set consists of 7130 documents along with abstractive summaries (called headnotes). The authors also sample 100 document-summary pairs for evaluation.

Dataset	Summary Type	Test Set Size	Training Set Size
BVA	Abs, segmented	20	92
IN-Abs	Abs, non-segmented	100	7030
UK-Abs	Abs, segmented	100	693

Table 3.1: The three datasets used in this work.

3.1.3 United Kingdom-Abstractive Dataset (UK-Abs)

The United Kingdom Supreme Court provides all case judgements ruled since 2009 on their website (https://www.supremecourt.uk/decided-cases/), and most are accompanied by an official press summary. (Shukla et al., 2022) collect a set of 793 case documents from 2009-2021, along with the press summary. 100 of these are reserved as the test set, and the remaining 693 instances constitute the training set. The summaries are divided into three segments – 'Background to the Appeal', 'Judgement', and 'Reasons for Judgement'.

In this work, we use the above described datasets in a cross-domain setting, by training supervised algorithms on the training set of one dataset, and testing on the test set of another. Table 3.1 summarizes the dataset used in this work.

3.2 Metrics for Text Summarization

In the realm of text summarization, multiple evaluation metrics are used, among which the ROUGE score and its variants are most common. ROUGE (Lin, 2004) stands for Recall-Oriented Understudy for Gisting Evaluation. It rates summaries based on the number of overlapping units (e.g. word pairs, n-grams or word sequences) with the reference summary. It has five different variants – ROUGE-N, ROUGE-L, ROUGE-W, ROUGE-S and ROUGE-SU. In this work, we focus only on ROUGE-N and ROUGE-L:

3.2.1 ROUGE-N: N-Gram Overlap

ROUGE-N is simply the n-gram recall between a generated summary and a set of reference summaries. It can be calculated as:

$$\text{ROUGE} - \text{N} = \frac{\sum_{S \in RefSummary} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in RefSummary} \sum_{gram_n \in S} Count(gram_n)}$$

Where S is a sentence, gram_n and $\text{count}_\text{match}$ is the number of N-grams matching in the generated and reference summary. n is the N-gram's length. The number of N-grams

increases with the addition of more reference summaries, which is intuitive because there can be multiple good summaries to a document. Furthermore, if a generated summary has matching n-grams with more than one reference, it gets a higher score. In this study, we use the ROUGE-1 and ROUGE-2 metrics.

3.2.2 ROUGE-L: Longest Common Subsequence

For two sequences $X = [x_1, x_2, ..., x_n]$ and $Y = [y_1, y_2, y_m]$, if there exists an increasing sequence $Z = [z_1, z_2, ..., z_k]$ of indices of X such that $x_{z_j} = y_j \ \forall j = 1, 2, ..., k$, X is a subsequence of Y (*Introduction to Algorithms* 1989). Therefore, the longest common subsequence (LCS) of two sequences X and Y is a common subsequence with the maximum length. LCS has historically been used as a similarity measure between two sentences (Melamed, 1995; Saggion et al., 2002). (Lin, 2004) propose an LCS-based F-measure for summary evaluation between a reference summary X of length m and a generated summary Y of length n given by:

$$R_{LCS} = \frac{LCS(X,Y)}{m}$$

$$P_{LCS} = \frac{LCS(X,Y)}{n}$$

$$F_{LCS} = \frac{(1+\beta^2)R_{LCS}P_{LCS}}{R_{LCS} + \beta^2 P_{LCS}}$$

Where $\beta = P_{LCS}/R_{LCS}$. LCS does not require consecutive matches, but rather in-sequence matches that reflect sentence level word order like n-grams. Moreover, it automatically includes the largest in-sequence n-gram match without any pre-defined n-gram length being specified. F_{LCS} is the ROUGE-L metric that we will use in this study.

3.2.3 BERTScore

ROUGE considers syntactic overlap between the hypothesis and reference statements by considering unigrams, bigrams, etc. However, it is limited by the presence of the exact word or phrase in both the hypothesis and reference, and hence does not decode the semantics of both statements. This is especially problematic for abstractive summarization, where the model aims to generate novel words and sentences, rather than *extract* sentences from the source document. BERTScore (Zhang, Kishore, et al., 2020) aims to understand the underlying semantics of the generated summary and compares it to the

reference. Figure 3.1 illustrates the computation of BERTScore. Both the reference (x) and hypothesis ($\hat{x}$) are passed through a pre-trained BERT model to obtain a contextual embedding for each word at the output end. Further, the pairwise cosine similarity of every word in the reference with every word in the hypothesis is computed. The most similar word from reference to the hypothesis is selected, and the precision, recall and F1-scores are calculated.

$$P_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} x_i^T \hat{x}_j \qquad R_{\text{BERT}} = \frac{1}{|x|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} x_i^T \hat{x}_j \qquad F_{\text{BERT}} = 2\frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

An optional importance weighting can also be done, using IDF scores, determined using a large corpus of offline text data. If a word has a high IDF score, it is an uncommon word across multiple documents. Thus, if such a word occurs both in the reference and hypothesis, it is worth considering in the final score. For our study, we report only the BERTScore F1-scores.

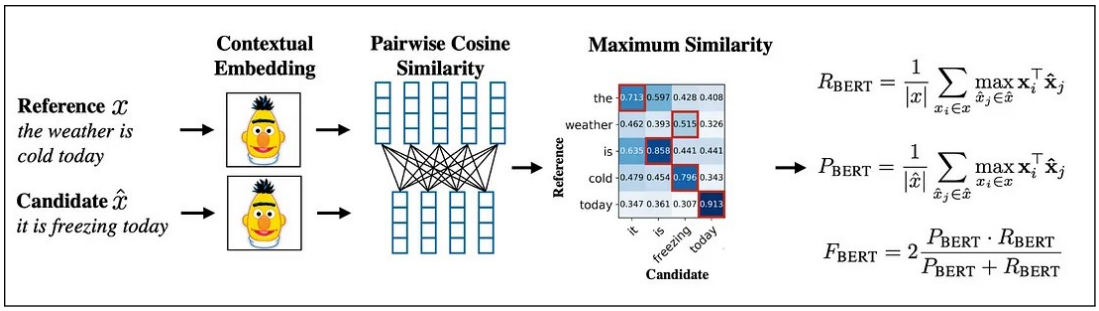

Figure 3.1: Computation of BERTScore. Given the reference x and candidate $\hat{x}$, compute the BERT embeddings and pairwise cosine similarity. the greedy matching is highlighted in red, and include the optional idf importance weighting.

Chapter 4

Experiments and Results

4.1 Experimental Setup

We perform an extensive evaluation across the BVA, UK-Abs and IN-Abs dataset. For the cross-domain evaluation study, we train and test our models on all three datasets. Concretely, we perform 9 evaluations per model.

We include a variety of models in our analysis: namely BART, BERT-BART, Legal-LED, Legal-Pegasus and SummaRuNNer among supervised models; and CaseSummarizer, LetSum, LexRank, LSA, LetSum, MMR and Reduction among unsupervised models. For BART, BERT-BART, Legal-LED and Legal-Pegasus, we also produce the results using the model without any fine-tuning. We do not perform any preprocessing on the text, or filtering in terms of rhetorics and segments.

4.1.1 Extractive Summarization Methods

The UIE models are implemented using the Sumy module in Python. We use the default settings as described in the module for our purpose. We follow (Shukla et al., 2022) in

Models	Model Type	Abbreviation
LexRank, LSA, Luhn, Reduction	Unsupervised, Domain-Agnostic, Extractive	UIE
CaseSummarizer, LetSum, MMR	Unsupervised, Domain-Specific, Extractive	ULE
SummaRuNNer	Supervised, Domain-Agnostic, tractive	SIE
BART, BERT-BART	Supervised, Domain-Agnostic, Abstractive	SIA
Legal-Pegasus, Legal-LED	Supervised, Domain-Specific, Abstractive	SLA

Table 4.1: The models used in this work.

their approach of using CaseSummarizer with a legal dictionary of common Indian legal terms, and MMR with only the baseline Maximum Marginal Relevance module due to the absence of a "Reasoning and Evidential Support" section. SummaRuNNer requires labelled training data, with sentences labelled as 1 if relevant to the summary, and 0 if not. Since our datasets are not natively in this format, we adopt the technique employed by (Narayan et al., 2018). For each sentence in the gold-standard summary, we select 3 sentences from the source document having the highest mean of R-1, R-2 and R-L scores. We label these sentences as 1, and the rest as 0.

4.1.2 Abstractive Summarization Models

Among short-document summarizers, we select Legal-Pegasus, which is already pre-trained on legal documents, and BART. We use the versions of these models available publicly in the HuggingFace library. Both these models have a maximum input token limit of 1024, which is significantly lesser than the average number of tokens in our datasets. Hence, we employ a chunking-based approach. Concretely, we divide the model into chunks, with each chunk having tokens lesser than the token limit of the model. Each chunk is then summarized by the model, and the resulting summaries are concatenated. To generate the fine-tuning data for these models, we use the method described by (Gidiotis et al., 2020). For a document-summary pair (d, s), when the document is split into n chunks $(d_1, d_2, ..., d_n)$, the reference summary mapped to these chunks must also be different. To generate suitable reference summaries, we map s to the most similar sentence in d. For this, we use Mean Cosine Similarity (MCS) (Ranasinghe et al., 2019), which calculates the mean of token embeddings (obtained using SBERT (Reimers et al., 2019)) which gives the sentence representation. We then compute the cosine similarity between two such sentence embeddings. Further, for every d_i we combine all sentences s which are mapped to any sentence in d_i and consider the resulting collection of sentences as s_i. The resulting (d_i, s_i) pairs are used for fine-tuning.

4.1.3 Adversarial Domain Adaptation

To improve the cross-domain performance of our models, we apply Adversarial Domain Adaptation introduced by (Ganin et al., 2016) and discussed in Section 2.4. We use the documents and reference summaries from the training set of the source domain, and only the documents from the testing set of the target domain. We also modify this approach to

further include silver-grade summaries, i.e. the generated summaries for the documents from the target domain as obtained from some summarization system. We use the encoder part of our models as the feature extractor, the decoder as the predictor, and construct a small feedforward neural network as the domain classifier. We use a schedule to control λ which is given by

$$\lambda = \frac{1}{1 + exp(-\gamma p)}$$

where t and T are the number of current training steps and total steps. γ requires careful tuning within [0.05,0.5] in increments of 0.05. This is illustrated by Figure 4.1, which shows the fluctuation of the R-1 score obtained by Legal-Pegasus under several settings.

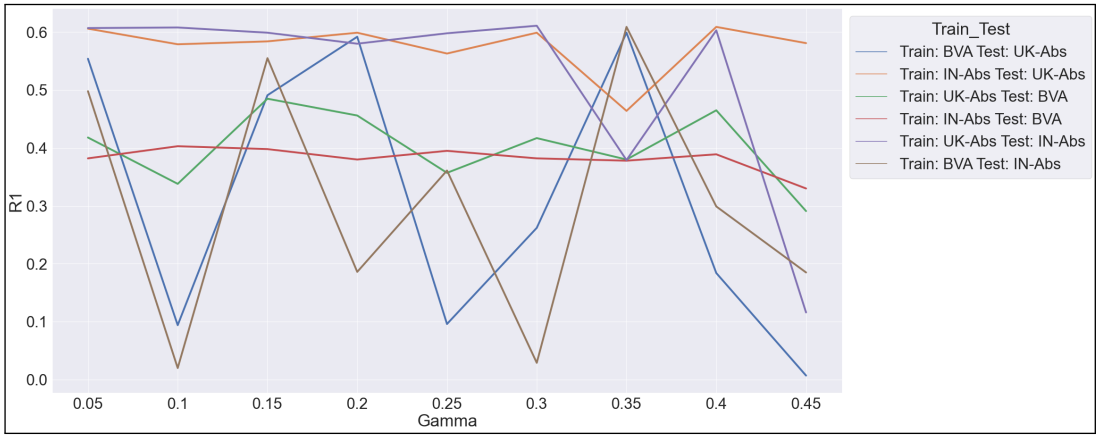

Figure 4.1: Variation of R-1 scores with γ when trained under several settings using Legal-Pegasus

4.2 Results and Analyses

In this section, we analyze the performance of different summarization models in a cross-domain setting. For the UK-Abs, IN-Abs and BVA datasets, Table 4.2, Table 4.3 and Table 4.4 report the results of several supervised and unsupervised; domain-agnostic and domain-specific; extractive and abstractive models. Table 4.5 contains the values of the stiffness and stableness metrics across all models. Table 4.6 and Table 4.7 contain the results for adversarial domain adaptation with and without using silver summaries for the target domain.

4.2.1 Cross-Domain Evaluation results

Among unsupervised, domain-agnostic, extractive (UAE) methods, Reduction performs the best almost always, with LexRank having a slightly higher BERTScore while testing

on UK-Abs and IN-Abs. However, CaseSummarizer and MMR outperform other UAE methods on UK-Abs and IN-Abs, because they are tailored to identify structures and semantics of legal documents. However, on BVA, UAE methods perform better than UDE methods.

Among supervised methods, SummaRuNNer has a significant performance gap compared to other supervised approaches, and even some unsupervised methods. This is potentially because SummaRuNNer is trained from scratch, unlike the other models which are already pre-trained on a large corpus. Legal-LED has a very poor performance without any fine-tuning (Vanilla case). This is expected, since it does not have a pre-trained decoder, and hence struggles to generate coherent summaries. On fine-tuning with BVA for cross-domain settings, the performance decreases further, possibly due to the small size of the BVA corpus and the domain shift between BVA and the other corpora being significant. It is worth noting that BART-based summarizers, which are made for short documents, consistently outperform Legal-LED, a long-document summarizer, by generating summaries by chunking. Legal-Pegasus too exhibits this behaviour, but falls back when tested on BVA.

Overall, in-domain testing almost always yields a better result than cross-domain testing, but cross-domain settings of supervised abstractive models still perform better than unsupervised methods for IN-Abs and UK-Abs. For BVA, unsupervised models perform better than cross-domain settings, which is likely due to the small size of the corpus. Thus, in the absence of labelled training data of a given jurisdiction, it is a better strategy to use fine-tune a model on documents from a different jurisdiction, rather than use unsupervised models.

Table 4.5 contains the stiffness and stableness measures for our supervised approaches. We see that the stableness measures for most models is very close to 1, which indicates that the performance gap between in-domain and cross-domain settings is quite small. Figure 4.2 shows the variation of stiffness and stableness for different metrics. Legal-Pegasus has the highest stiffness and stableness score across all metrics, indicating that it not only performs better, but is also more robust. SummaRuNNer has a much lesser stiffness score than other models, but has the second-best stableness score, making it more robust than Legal-LED and BART-based approaches.

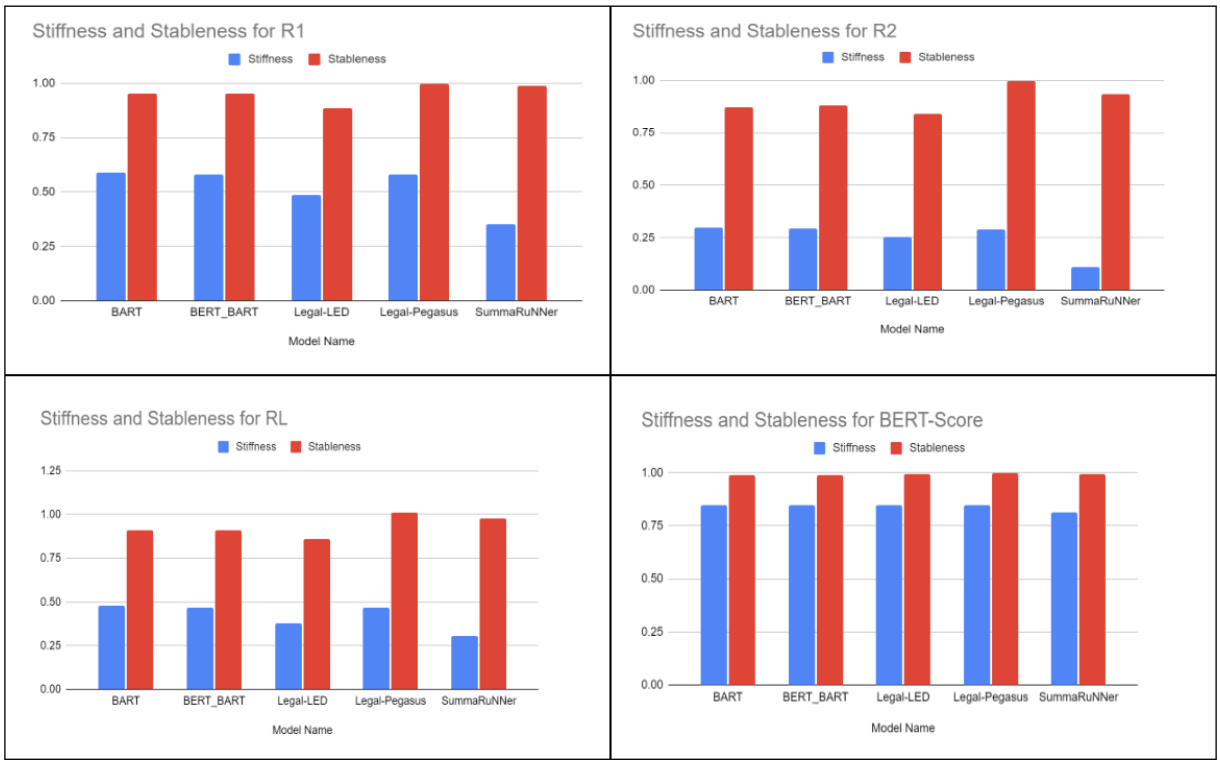

Figure 4.2: Cross-Domain Evaluation results. A high stiffness indicates high cross-domain performance, and high stableness indicates robustness of the model

4.2.2 Evaluation of Domain Adaptation methods

Table 4.6 and Table 4.7 contain the results of applying Adversarial Domain Adaptation (ADA). We employ this technique only on the two best-performing models from the cross-domain evaluation setting: Legal-Pegasus and BART. For Legal-Pegasus, applying ADA doesn't always lead to a performance increase. This behaviour can be explained by the small sizes of all test datasets (100 for IN-Abs and UK-Abs, and 20 for BVA), leading to less adaptation.

However, when silver-grade summaries are introduced, the performance almost certainly improves. While testing on BVA, ADA with silver summaries attain the best performance, with summaries generated by MMR being the best, and achieves a 14% improvement when transferred from IN-Abs. However, the improvement is not consistent across datasets, and when transferred from UK-Abs and tested on IN-Abs, the non-DA approach performs better.

While using BART as the base model, introducing domain adaptation always achieves similar performance or improves upon the non-DA case, with a 51% improvement in R-

2 score when transferred from IN-Abs and tested on BVA using MMR-generated silver summaries. This indicates the power of using domain adaptation for our purpose.

4.3 Conclusion and Future Work

We study the cross-domain summarization aptitude of several models: supervised and unsupervised, domain-agnostic and domain-specific, extractive and abstractive. We find the in-domain performance to almost always exceed the cross-domain performance, but in the absence of labelled training data from a certain jurisdiction, it is worth using the labelled data from other jurisdictions rather than employing unsupervised techniques. Furthermore, we investigate the usefulness of employing domain adaptation to increase the cross-domain performance of attention-based supervised algorithms, with the introduction of silver-grade summaries to further boost performance. In the future, we plan on experimenting with more models like BERT-Abs and BERT-Ext, and legal-domain specific extractive models like Gist. We also plan on using parameter-efficient finetuning methods like adapter modules (Houlsby et al., 2019), BitFit (Ben Zaken et al., 2022) and LMFit (Frankle et al., 2020), which have also shown some promise in domain adaptation. Another potential goal would be to expand the cross-domain aptitude of these modules to other languages as well,

Model	Train Dataset	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
Unsupervised, Domain-Agnostic, Extractive Models					
LexRank	None	0.483	0.225	0.427	**0.836**
LSA	None	0.419	0.159	0.361	0.823
Luhn	None	0.501	0.194	0.435	0.828
Reduction	None	0.551	0.228	0.484	0.829
Unsupervised, Domain-Specific, Extractive Models					
CaseSummarizer	None	**0.596**	**0.251**	0.519	0.835
LetSum	None	0.4	0.139	0.36	0.772
MMR	None	0.595	0.248	**0.567**	0.834
Supervised, Domain-Agnostic, Extractive Models					
SummaRuNNer	BVA	0.43	0.196	0.378	0.848
SummaRuNNer	IN-Abs	0.487	0.225	0.43	0.844
SummaRuNNer	UK-Abs	0.459	0.223	0.408	0.856
Supervised, Domain-Agnostic, Abstractive Models					
BART	Vanilla	0.602	0.243	0.517	0.836
BART	BVA	0.598	0.242	0.519	0.836
BART	IN-Abs	0.615	0.255	0.532	0.837
BART	UK-Abs	**0.627**	**0.272**	**0.543**	**0.844**
BERT-BART	Vanilla	0.586	0.227	0.501	0.837
BERT-BART	BVA	0.59	0.235	0.508	0.838
BERT-BART	IN-Abs	0.599	0.244	0.515	0.837
BERT-BART	UK-Abs	0.615	0.26	0.529	0.845
Supervised, Domain-Specific, Abstractive Models					
Legal-LED	Vanilla	0.272	0.089	0.24	0.816
Legal-LED	BVA	0.265	0.116	0.233	0.822
Legal-LED	IN-Abs	0.556	0.239	0.479	0.846
Legal-LED	UK-Abs	0.541	0.245	0.467	0.848
Legal-Pegasus	Vanilla	0.59	0.242	0.517	0.836
Legal-Pegasus	BVA	0.596	0.247	0.522	0.836
Legal-Pegasus	IN-Abs	0.617	0.259	0.533	0.834
Legal-Pegasus	UK-Abs	0.624	0.271	0.541	0.838

Table 4.2: Results on the UK-Abs Dataset. The best results across each metric for unsupervised and supervised methods is emphasised. The best score achieved by each category is highlighted in gray

Model	Train Dataset	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
Unsupervised, Domain-Agnostic, Extractive Models					
LexRank	None	0.523	0.286	0.459	**0.85**
LSA	None	0.479	0.236	0.41	0.841
Luhn	None	0.522	0.27	0.454	0.843
Reduction	None	0.537	**0.300**	0.473	0.847
Unsupervised, Domain-Specific, Extractive Models					
CaseSummarizer	None	**0.599**	**0.300**	0.517	0.836
LetSum	None	0.405	0.144	0.362	0.797
MMR	None	0.581	0.296	**0.548**	0.847
Supervised, Domain-Agnostic, Extractive Models					
SummaRuNNer	BVA	0.438	0.199	0.374	0.832
SummaRuNNer	IN-Abs	0.536	0.303	0.471	0.849
SummaRuNNer	UK-Abs	0.473	0.23	0.41	0.839
Supervised, Domain-Agnostic, Abstractive Models					
BART	Vanilla	0.57	0.268	0.486	0.836
BART	BVA	0.581	0.277	0.503	0.84
BART	IN-Abs	0.625	0.337	**0.548**	**0.851**
BART	UK-Abs	0.608	0.311	0.522	0.847
BERT-BART	Vanilla	0.566	0.266	0.482	0.836
BERT-BART	BVA	0.58	0.279	0.501	0.839
BERT-BART	IN-Abs	0.62	0.33	0.541	0.851
BERT-BART	UK-Abs	0.6	0.305	0.513	0.846
Supervised, Domain-Specific, Abstractive Models					
Legal-LED	Vanilla	0.342	0.108	0.292	0.815
Legal-LED	BVA	0.337	0.156	0.289	0.827
Legal-LED	IN-Abs	0.555	0.301	0.48	0.855
Legal-LED	UK-Abs	0.493	0.231	0.418	0.838
Legal-Pegasus	Vanilla	0.601	0.302	0.522	0.845
Legal-Pegasus	BVA	0.601	0.303	0.522	0.845
Legal-Pegasus	IN-Abs	**0.629**	**0.342**	0.547	0.85
Legal-Pegasus	UK-Abs	0.615	0.323	0.532	0.847

Table 4.3: Results on the IN-Abs Dataset. The best results across each metric for unsupervised and supervised methods is emphasised. The best score achieved by each category is highlighted in gray

Model	Train Dataset	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
Unsupervised, Domain-Agnostic, Extractive Models					
LexRank	None	**0.546**	0.341	0.356	0.862
LSA	None	0.516	0.281	0.308	0.854
Luhn	None	0.53	0.34	0.363	0.86
Reduction	None	0.534	**0.373**	**0.394**	**0.865**
Unsupervised, Domain-Specific, Extractive Models					
CaseSummarizer	None	0.436	0.181	0.346	0.829
LetSum	None	0.391	0.156	0.298	0.807
MMR	None	0.514	0.267	0.384	0.825
Supervised, Domain-Agnostic, Extractive Models					
SummaRuNNer	BVA	0.516	0.423	0.453	0.878
SummaRuNNer	IN-Abs	0.455	0.273	0.327	0.846
SummaRuNNer	UK-Abs	0.497	0.368	0.405	0.863
Supervised, Domain-Agnostic, Abstractive Models					
BART	Vanilla	0.482	0.301	0.344	0.834
BART	BVA	**0.605**	**0.435**	**0.472**	**0.881**
BART	IN-Abs	0.481	0.226	0.306	0.837
BART	UK-Abs	0.555	0.334	0.34	0.861
BERT-BART	Vanilla	0.492	0.307	0.349	0.835
BERT-BART	BVA	0.602	0.427	0.465	0.88
BERT-BART	IN-Abs	0.482	0.229	0.308	0.839
BERT-BART	UK-Abs	0.552	0.334	0.342	0.86
Supervised, Domain-Specific, Abstractive Models					
Legal-LED	Vanilla	0.447	0.174	0.232	0.833
Legal-LED	BVA	0.552	0.353	0.379	0.85
Legal-LED	IN-Abs	0.541	0.315	0.339	0.859
Legal-LED	UK-Abs	0.547	0.323	0.321	0.859
Legal-Pegasus	Vanilla	0.493	0.246	0.296	0.847
Legal-Pegasus	BVA	0.503	0.264	0.305	0.85
Legal-Pegasus	IN-Abs	0.518	0.292	0.336	0.854
Legal-Pegasus	UK-Abs	0.528	0.308	0.343	0.856

Table 4.4: Results on the BVA Dataset. The best results across each metric for unsupervised and supervised methods is emphasised. The best score achieved by each category is highlighted in gray

Model Name	Metric	Stiffness	Stableness
BART	BERTScore	0.848	0.988
BERT-BART	BERTScore	0.848	0.988
Legal-LED	BERTScore	0.845	0.993
Legal-Pegasus	BERTScore	0.846	0.999
SummaRuNNer	BERTScore	0.813	0.995
BART	ROUGE-1	0.588	0.95
BERT-BART	ROUGE-1	0.582	0.95
Legal-LED	ROUGE-1	0.487	0.887
Legal-Pegasus	ROUGE-1	0.581	0.995
SummaRuNNer	ROUGE-1	0.353	0.987
BART	ROUGE-2	0.299	0.873
BERT-BART	ROUGE-2	0.294	0.881
Legal-LED	ROUGE-2	0.253	0.838
Legal-Pegasus	ROUGE-2	0.29	0.997
SummaRuNNer	ROUGE-2	0.11	0.933
BART	ROUGE-L	0.476	0.908
BERT-BART	ROUGE-L	0.469	0.912
Legal-LED	ROUGE-L	0.378	0.86
Legal-Pegasus	ROUGE-L	0.465	1.011
SummaRuNNer	ROUGE-L	0.304	0.976

Table 4.5: Cross-Domain Evaluation results for supervised models. Stiffness and Stableness are calculated as described in Section 2.3

Model	Train Dataset	Test Dataset	R1	R2	RL	BS	Gamma
Legal-Pegasus	IN-Abs	BVA	0.518	0.292	0.434	0.854	NA
Legal-Pegasus_CaseSumm	IN-Abs	BVA	0.538	0.325	0.459	0.856	0.15
Legal-Pegasus_CaseSumm	IN-Abs	BVA	0.548	0.306	0.455	0.855	0.4
Legal-Pegasus_DA	IN-Abs	BVA	0.403	0.154	0.326	0.818	0.1
Legal-Pegasus_MMR	IN-Abs	BVA	0.546	0.335	0.465	0.862	0.2
Legal-Pegasus	UK-Abs	BVA	0.528	0.308	0.451	0.856	NA
Legal-Pegasus_CaseSumm	UK-Abs	BVA	0.543	0.337	0.461	0.859	0.4
Legal-Pegasus_DA	UK-Abs	BVA	0.485	0.242	0.394	0.842	0.15
Legal-Pegasus_MMR	UK-Abs	BVA	0.546	0.318	0.466	0.858	0.1
Legal-Pegasus_MMR	UK-Abs	BVA	0.546	0.329	0.469	0.861	0.25
Legal-Pegasus_MMR	UK-Abs	BVA	0.546	0.329	0.464	0.86	0.3
Legal-Pegasus	BVA	IN-Abs	0.601	0.303	0.522	0.845	NA
Legal-Pegasus_CaseSumm	BVA	IN-Abs	0.598	0.295	0.513	0.837	0.2
Legal-Pegasus_DA	BVA	IN-Abs	0.609	0.301	0.523	0.843	0.35
Legal-Pegasus_MMR	BVA	IN-Abs	0.6	0.3	0.515	0.843	0.3
Legal-Pegasus	UK-Abs	IN-Abs	0.615	0.323	0.532	0.847	NA
Legal-Pegasus_CaseSumm	UK-Abs	IN-Abs	0.592	0.287	0.506	0.833	0.15
Legal-Pegasus_DA	UK-Abs	IN-Abs	0.611	0.297	0.524	0.842	0.3
Legal-Pegasus_MMR	UK-Abs	IN-Abs	0.608	0.309	0.526	0.845	0.25
Legal-Pegasus	BVA	UK-Abs	0.596	0.247	0.522	0.836	NA
Legal-Pegasus_CaseSumm	BVA	UK-Abs	0.609	0.245	0.524	0.83	0.25
Legal-Pegasus_DA	BVA	UK-Abs	0.599	0.235	0.517	0.83	0.35
Legal-Pegasus_MMR	BVA	UK-Abs	0.6	0.241	0.517	0.831	0.1
Legal-Pegasus_MMR	BVA	UK-Abs	0.6	0.243	0.518	0.832	0.25
Legal-Pegasus	IN-Abs	UK-Abs	0.617	0.259	0.533	0.834	NA
Legal-Pegasus_CaseSumm	IN-Abs	UK-Abs	0.622	0.253	0.536	0.831	0.45
Legal-Pegasus_DA	IN-Abs	UK-Abs	0.609	0.235	0.528	0.825	0.4
Legal-Pegasus_MMR	IN-Abs	UK-Abs	0.617	0.249	0.535	0.831	0.4

Table 4.6: Results of using Adversarial Domain Adaptation on Legal-Pegasus. ***_DA** indicates finetuning the model using Gradient Reversal Layers. ***_CaseSumm/MMR** indicates finetuning the model using GRLs along with silver-summaries for the target domain, ie, summaries of the target domain data as generated by CaseSummarizer or MMR. Best results in each category are highlighted in grey.

Model	Train Dataset	Test Dataset	R1	R2	RL	BS	Gamma
BART	IN-Abs	BVA	0.481	0.226	0.306	0.837	None
BART_DA	IN-Abs	BVA	0.52	0.292	0.434	0.853	0.1
BART_DA_MMR	IN-Abs	BVA	0.551	0.342	0.472	0.861	0.2
BART	UK-Abs	BVA	0.555	0.334	0.34	0.861	None
BART_DA	UK-Abs	BVA	0.553	0.336	0.46	0.866	0.05
BART_DA_MMR	UK-Abs	BVA	0.553	0.342	0.472	0.864	0.1
BART	BVA	IN-Abs	0.581	0.277	0.503	0.84	None
BART_DA	BVA	IN-Abs	0.576	0.275	0.507	0.84	0.05
BART_DA_MMR	BVA	IN-Abs	0.587	0.279	0.499	0.841	0.05
BART	UK-Abs	IN-Abs	0.608	0.311	0.522	0.847	None
BART_DA	UK-Abs	IN-Abs	0.607	0.31	0.521	0.847	0.05
BART_DA_MMR	UK-Abs	IN-Abs	0.597	0.29	0.507	0.843	0.05
BART	BVA	UK-Abs	0.598	0.242	0.519	0.836	None
BART_DA	BVA	UK-Abs	0.593	0.235	0.524	0.837	0.05
BART_DA_MMR	BVA	UK-Abs	0.601	0.242	0.526	0.836	0.05
BART	IN-Abs	UK-Abs	0.615	0.255	0.532	0.837	None
BART_DA	IN-Abs	UK-Abs	0.62	0.264	0.536	0.837	0.05
BART_DA_MMR	IN-Abs	UK-Abs	0.604	0.248	0.527	0.836	0.15

Table 4.7: Results of using Adversarial Domain Adaptation on BART. ***_DA** indicates finetuning the model using Gradient Reversal Layers. ***_MMR** indicates finetuning the model using GRLs along with silver-summaries for the target domain, ie, summaries of the target domain data as generated by MMR. Best results in each category are highlighted in gray.

References

Bajaj, Ahsaas, Pavitra Dangati, Kalpesh Krishna, Pradhiksha Ashok Kumar, Rheeya Uppaal, Bradford Windsor, Eliot Brenner, Dominic Dotterrer, Rajarshi Das, and Andrew McCallum (2021). *Long Document Summarization in a Low Resource Setting using Pretrained Language Models.*

Barzilay, Regina, Kathleen R. McKeown, and Michael Elhadad (June 1999). "Information Fusion in the Context of Multi-Document Summarization". In: *Proceedings of the 37th Annual Meeting of the Association for Computational Linguistics.* College Park, Maryland, USA: Association for Computational Linguistics, pp. 550–557.

Beltagy, Iz, Matthew E. Peters, and Arman Cohan (2020). *Longformer: The Long-Document Transformer.*

Ben Zaken, Elad, Yoav Goldberg, and Shauli Ravfogel (May 2022). "BitFit: Simple Parameter-efficient Fine-tuning for Transformer-based Masked Language-models". In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers).* Dublin, Ireland: Association for Computational Linguistics, pp. 1–9.

Carbonell, Jaime and Jade Goldstein (1998). "The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries". In: *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval.* SIGIR '98. Melbourne, Australia: Association for Computing Machinery, pp. 335–336.

Carletta, Jean, Simone Ashby, Sebastien Bourban, Mike Flynn, Mael Guillemot, Thomas Hain, Jaroslav Kadlec, Vasilis Karaiskos, Wessel Kraaij, Melissa Kronenthal, Guillaume Lathoud, Mike Lincoln, Agnes Lisowska, Iain McCowan, Wilfried Post, Dennis Reidsma, and Pierre Wellner (2006). "The AMI Meeting Corpus: A Pre-announcement". In: *Machine Learning for Multimodal Interaction.* Ed. by Steve Renals and Samy Bengio. Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 28–39.

Chen, Francine and Yan-Ying Chen (July 2019). "Adversarial Domain Adaptation Using Artificial Titles for Abstractive Title Generation". In: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Florence, Italy: Association for Computational Linguistics, pp. 2197–2203.

Chen, Yiran, Pengfei Liu, Ming Zhong, Zi-Yi Dou, Danqing Wang, Xipeng Qiu, and Xuanjing Huang (Nov. 2020). "CDEvalSumm: An Empirical Study of Cross-Dataset Evaluation for Neural Summarization Systems". In: *Findings of the Association for Computational Linguistics: EMNLP 2020*. Online: Association for Computational Linguistics, pp. 3679–3691.

Cheng, Jianpeng and Mirella Lapata (2016). *Neural Summarization by Extracting Sentences and Words*.

Erkan, G. and D. R. Radev (Dec. 2004). "LexRank: Graph-based Lexical Centrality as Salience in Text Summarization". In: *Journal of Artificial Intelligence Research* 22, pp. 457–479.

Farzindar, Atefeh and Guy Lapalme (Nov. 2004). "LetSum, an automatic Legal Text Summarizing system". In: McMaster University, Hamilton, Ontario, Canada: THE FACE OF TEXT conference (Computer Assisted Text Analysis in the Humanities).

Frankle, Jonathan, David J. Schwab, and Ari S. Morcos (2020). "Training BatchNorm and Only BatchNorm: On the Expressive Power of Random Features in CNNs". In: *CoRR* abs/2003.00152.

Ganin, Yaroslav, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky (2016). *Domain-Adversarial Training of Neural Networks*.

Gidiotis, Alexios and Grigorios Tsoumakas (2020). *A Divide-and-Conquer Approach to the Summarization of Long Documents*.

Gong, Yihong and Xin Liu (2001). "Generic Text Summarization Using Relevance Measure and Latent Semantic Analysis". In: *Proceedings of the 24th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR '01. New Orleans, Louisiana, USA: Association for Computing Machinery, pp. 19–25.

Harris, Zellig S. (1954). "Distributional Structure". In: *WORD* 10.2-3, pp. 146–162.

Hochreiter, Sepp and Jürgen Schmidhuber (1997). "Long short-term memory". In: *Neural computation* 9.8, pp. 1735–1780.

Houlsby, Neil, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly (2019). "Parameter-Efficient Transfer Learning for NLP". In: *CoRR* abs/1902.00751.

Introduction to Algorithms (1989).

Jing, Hongyan (Apr. 2000). "Sentence Reduction for Automatic Text Summarization". In: *Sixth Applied Natural Language Processing Conference*. Seattle, Washington, USA: Association for Computational Linguistics, pp. 310–315.

Klema, V. and A. Laub (1980). "The singular value decomposition: Its computation and some applications". In: *IEEE Transactions on Automatic Control* 25.2, pp. 164–176.

Lewis, Mike, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer (2019). *BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension.*

Lin, Chin-Yew (July 2004). "ROUGE: A Package for Automatic Evaluation of Summaries". In: *Text Summarization Branches Out*. Barcelona, Spain: Association for Computational Linguistics, pp. 74–81.

Liu, Yang and Mirella Lapata (2019). *Text Summarization with Pretrained Encoders.*

Luhn, H. P. (Apr. 1958). "The Automatic Creation of Literature Abstracts". In: 2.2, pp. 159–165.

Melamed, I. Dan (1995). "Automatic Evaluation and Uniform Filter Cascades for Inducing N-Best Translation Lexicons". In: *Third Workshop on Very Large Corpora.*

Mikolov, Tomas, Kai Chen, Greg Corrado, and Jeffrey Dean (2013). *Efficient Estimation of Word Representations in Vector Space.*

Mishra, Rashmi, Jiantao Bian, Marcelo Fiszman, Charlene R Weir, Siddhartha Jonnalagadda, Javed Mostafa, and Guilherme Del Fiol (Dec. 2014). "Text summarization in the biomedical domain: a systematic review of recent research". In: *Journal of biomedical informatics* 52, pp. 457–467.

Naik, Aakanksha and Carolyn Rose (July 2020). "Towards Open Domain Event Trigger Identification using Adversarial Domain Adaptation". In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Online: Association for Computational Linguistics, pp. 7618–7624.

Nallapati, Ramesh, Feifei Zhai, and Bowen Zhou (2016). *SummaRuNNer: A Recurrent Neural Network based Sequence Model for Extractive Summarization of Documents.*

Nallapati, Ramesh, Bowen Zhou, Cicero Nogueira dos santos, Caglar Gulcehre, and Bing Xiang (2016). "Abstractive Text Summarization Using Sequence-to-Sequence RNNs and Beyond". In.

Narayan, Shashi, Shay B. Cohen, and Mirella Lapata (June 2018). "Ranking Sentences for Extractive Summarization with Reinforcement Learning". In: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*. New Orleans, Louisiana: Association for Computational Linguistics, pp. 1747–1759.

Ouyang, Jessica, Serina Chang, and Kathy McKeown (Apr. 2017). "Crowd-Sourced Iterative Annotation for Narrative Summarization Corpora". In: *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*. Valencia, Spain: Association for Computational Linguistics, pp. 46–51.

Page, Lawrence, Sergey Brin, Rajeev Motwani, and Terry Winograd (Nov. 1999). *The PageRank Citation Ranking: Bringing Order to the Web.* Technical Report 1999-66. Previous number = SIDL-WP-1999-0120. Stanford InfoLab.

Pennington, Jeffrey, Richard Socher, and Christopher Manning (Oct. 2014). "GloVe: Global Vectors for Word Representation". In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Doha, Qatar: Association for Computational Linguistics, pp. 1532–1543.

Polsley, Seth, Pooja Jhunjhunwala, and Ruihong Huang (Dec. 2016). "CaseSummarizer: A System for Automated Summarization of Legal Texts". In: *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: System Demonstrations*. Osaka, Japan: The COLING 2016 Organizing Committee, pp. 258–262.

Raffel, Colin, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu (2019). *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer.*

Ranasinghe, Tharindu, Constantin Orasan, and Ruslan Mitkov (Sept. 2019). "Enhancing Unsupervised Sentence Similarity Methods with Deep Contextualised Word Representations". In: *Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2019)*. Varna, Bulgaria: INCOMA Ltd., pp. 994–1003.

REFERENCES

Reimers, Nils and Iryna Gurevych (2019). *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.*

Saggion, Horacio, Dragomir Radev, Simone Teufel, and Wai Lam (2002). "Meta-evaluation of Summaries in a Cross-lingual Environment using Content-based Metrics". In: *COLING 2002: The 19th International Conference on Computational Linguistics.*

Sairam, Naren (2021a). *https://huggingface.co/nsi319/legal-led-base-16384.*

Sairam, Naren (2021b). *https://huggingface.co/nsi319/legal-pegasus.*

See, Abigail, Peter J. Liu, and Christopher D. Manning (2017). *Get To The Point: Summarization with Pointer-Generator Networks.*

Shukla, Abhay, Paheli Bhattacharya, Soham Poddar, Rajdeep Mukherjee, Kripabandhu Ghosh, Pawan Goyal, and Saptarshi Ghosh (2022). *Legal Case Document Summarization: Extractive and Abstractive Methods and their Evaluation.*

Vinyals, Oriol, Meire Fortunato, and Navdeep Jaitly (2015). "Pointer Networks". In: *Advances in Neural Information Processing Systems.* Ed. by C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett. Vol. 28. Curran Associates, Inc.

Zhang, Jingqing, Yao Zhao, Mohammad Saleh, and Peter J. Liu (2019). *PEGASUS: Pre-training with Extracted Gap-sentences for Abstractive Summarization.*

Zhang, Tianyi, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi (2020). *BERTScore: Evaluating Text Generation with BERT.*

Zhong, Linwu, Ziyi Zhong, Zinian Zhao, Siyuan Wang, Kevin D. Ashley, and Matthias Grabmair (2019). "Automatic Summarization of Legal Decisions Using Iterative Masking of Predictive Sentences". In: *Proceedings of the Seventeenth International Conference on Artificial Intelligence and Law.* ICAIL '19. Montreal, QC, Canada: Association for Computing Machinery, pp. 163–172.